

PROBLEMA 5

a) $f(n) = \Theta(g(n))$ y $g(n) = \Theta(h(n)) \rightarrow h(n) = \Theta(f(n))$

Es verdadera por transitividad

$$c_1 g(n) \leq f(n) \leq c_2 g(n) \quad \text{y} \quad c_3 h(n) \leq g(n) \leq c_4 h(n)$$

$$c_1(c_3 h(n)) \leq f(n) \leq c_2 c_4 h(n) \Rightarrow f(n) = \Theta(h(n)) //$$

b) $f(n) = O(g(n))$ y $g(n) = O(h(n)) \rightarrow h(n) = \Omega(f(n))$

Verdadera por transitividad

c) $f(n) = \Theta(n^2)$

hacer hashing: $(j-1)$

$$\sum_{i=0}^{n-1} \sum_{j=i+1}^n (j-1) = \sum_{k=1}^n (n-k+1) \cdot k = \Theta(n^3) \therefore \text{Es falso} //$$